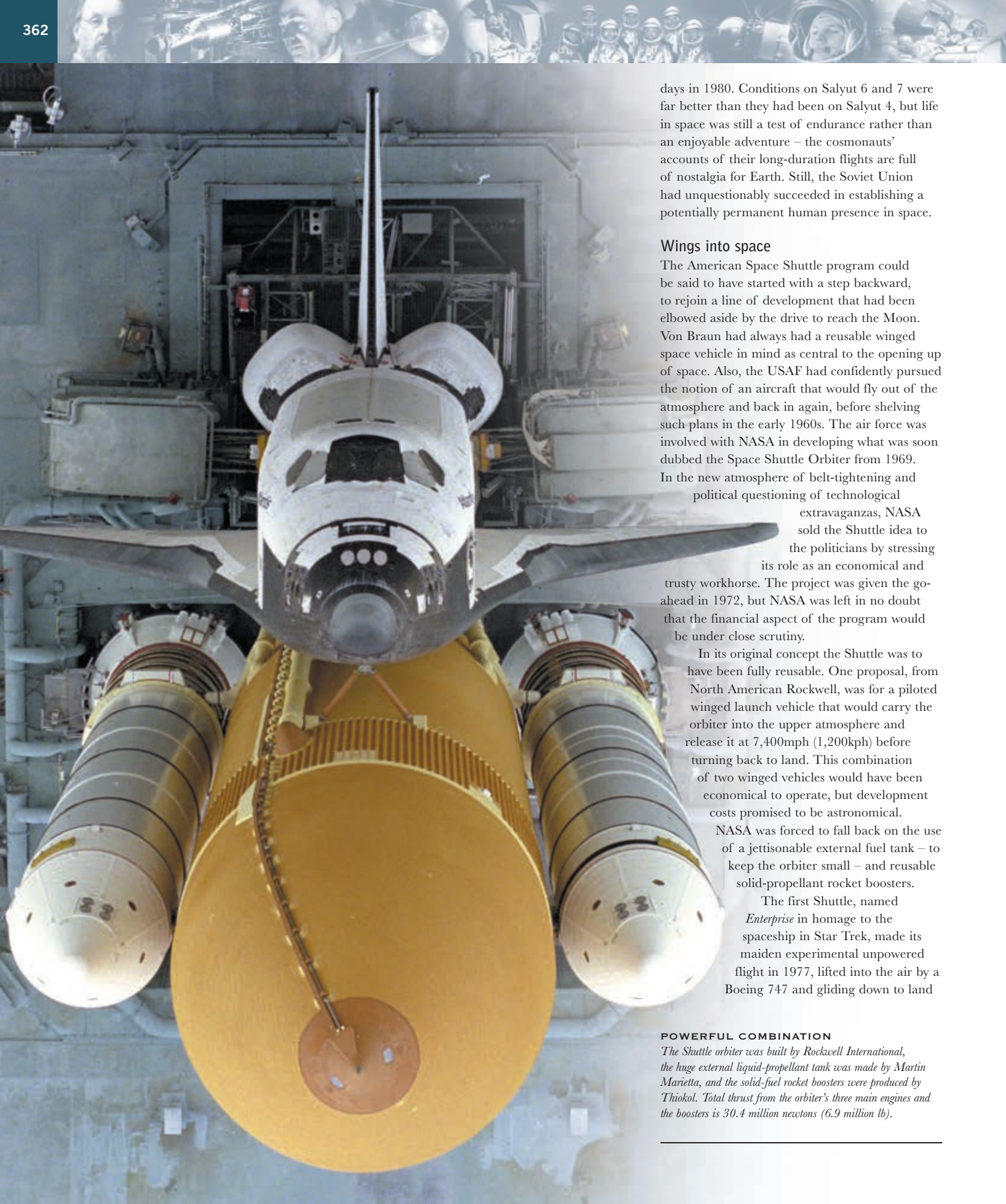




days in 1980. Conditions on Salyut 6 and 7 were far better than they had been on Salyut 4, but life in space was still a test of endurance rather than an enjoyable adventure – the cosmonauts' accounts of their long-duration flights are full of nostalgia for Earth. Still, the Soviet Union had unquestionably succeeded in establishing a potentially permanent human presence in space.

Wings into space

The American Space Shuttle program could be said to have started with a step backward, to rejoin a line of development that had been elbowed aside by the drive to reach the Moon. Von Braun had always had a reusable winged space vehicle in mind as central to the opening up of space. Also, the USAF had confidently pursued the notion of an aircraft that would fly out of the atmosphere and back in again, before shelving such plans in the early 1960s. The air force was involved with NASA in developing what was soon dubbed the Space Shuttle Orbiter from 1969. In the new atmosphere of belt-tightening and political questioning of technological

extravaganzas, NASA sold the Shuttle idea to the politicians by stressing its role as an economical and trusty workhorse. The project was given the go-ahead in 1972, but NASA was left in no doubt that the financial aspect of the program would be under close scrutiny.

In its original concept the Shuttle was to have been fully reusable. One proposal, from North American Rockwell, was for a piloted winged launch vehicle that would carry the orbiter into the upper atmosphere and release it at 7,400mph (1,200kph) before turning back to land. This combination of two winged vehicles would have been economical to operate, but development costs promised to be astronomical.

NASA was forced to fall back on the use of a jettisonable external fuel tank – to keep the orbiter small – and reusable solid-propellant rocket boosters.

The first Shuttle, named *Enterprise* in homage to the spaceship in *Star Trek*, made its maiden experimental unpowered flight in 1977, lifted into the air by a Boeing 747 and gliding down to land

POWERFUL COMBINATION

The Shuttle orbiter was built by Rockwell International, the huge external liquid-propellant tank was made by Martin Marietta, and the solid-fuel rocket boosters were produced by Thiokol. Total thrust from the orbiter's three main engines and the boosters is 30.4 million newtons (6.9 million lb).